

SPARE: The Sparse-view Reconstruction Challenge for 4D-CBCT

Instructions for using the datasets

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1 Introduction

This section gives a general overview of the SPARE challenge. For more details, please visit the SPARE challenge webpage on the ACRF Image X Institute website:

<https://image-x.sydney.edu.au/spare-challenge/>

1.1 What is the SPARE challenge?

The SPARE challenge stands for Sparse-view Reconstruction Challenge for Four-dimensional Cone-beam CT (4D CBCT). The aim is to systematically investigate the efficacy of various algorithms for 4D CBCT reconstruction from a one-minute scan. The goal of the SPARE challenge is to explore the possibility of high quality 4D CBCT while sparing patients the additional scan time and imaging dose that is currently used in the clinics.

4D CBCT allows the verification of tumor motion for thoracic patients immediately before every radiotherapy session. Currently a clinical 4D CBCT scan takes 2-4 minutes, as compared to one minute for a 3D CBCT scan. The recent emergence of image reconstruction algorithms that account for data sparsity offers new possibility for 4D CBCT reconstruction using a shorter scan. Specifically, the ability to reconstruct 4D CBCT images from a one-minute scan would reduce scan time and imaging dose. Additionally, every radiotherapy centre would gain access to 4D CBCT using existing 3D CBCT scanning protocol.

1.2 Host

The SPARE challenge is led by Dr Andy Shieh and Prof Paul Keall at the ACRF Image X Institute, a research institute under The University of Sydney. Collaborators who have contributed to the datasets include A/Prof Xun Jia, Miss Yesenia Gonzalez, and Mr Bin Li from the University of Texas Southwestern Medical Center, and Dr Simon Rit from the Creatis Medical Imaging Research Center.

1.3 Workflow

Participants will receive CBCT projection data either acquired clinically or simulated. Prior CT data and training sets will also be provided. Using these datasets, participants will apply their reconstruction algorithms and submit the final reconstructions to the host within a timeframe of two months. The participants are allowed to apply any

processing and algorithms to reconstruct the 4D CBCT images, including scatter correction, projection smoothing, iterative correction, etc.

Together with the reconstruction submission, the participants will be asked to include descriptions of the methods they have applied. The host will evaluate the performance of each algorithm using a set of ground truths that are not available to the participants.

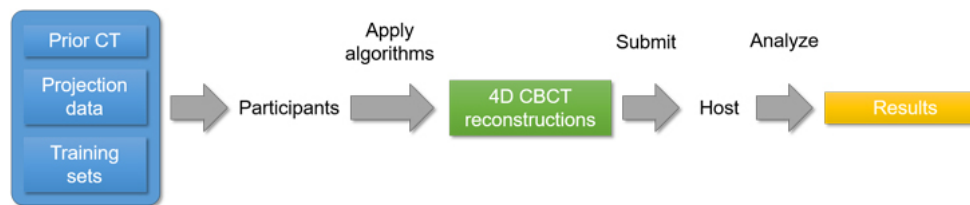


Figure 1. The workflow of the SPARE challenge.

1.4 Contact

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2 Datasets

This section gives an overview of the types of datasets involved in the challenge study, and how the participants can use them in the reconstruction.

If a password is required to download or unzip any files, it is:

#MakeIt1Minute!

2.1 Types of data

The SPARE challenge consists of three types of datasets:

1. Monte Carlo datasets

Projections simulated from 4D-CT volumes. This is the only dataset where an actual ground truth exists for every scan. Scatter and quantum noise were included in the projections depending on the simulated cases (no scatter/with scatter/low dose; see 2.6 for more details).

2. Clinical Varian datasets

4-8-minute patient scans acquired from a Varian Clinac for research purpose. Participants are given the down-sampled subsets equivalent to one-minute scans. The FDK reconstruction from the full projection set is used as the reference for performance assessment.

3. Clinical Elekta datasets

3-minute patient scans acquired on an Elekta Synergy in clinical practice. Similarly to the Clinical Varian datasets, participants are given the down-sampled sets, while the FDK reconstruction from the full set is used as the reference.

Quantitative analysis of the algorithm performance is only possible with the Monte Carlo datasets due to the availability of ground truths. The Clinical Varian and Elekta datasets are used to assess whether the performance on the Monte Carlo datasets can be generalized to actual clinical scans.

2.2 Folder structure and naming

Generally, the folder hierarchy is:

[Data type] - [Patient number] - [Scan number]

The only exception is for Monte Carlo datasets, where the training and validation sets are separate from a higher level:

MonteCarloDatasets - [Training/validation] - [Patient number] -
[Scan number]

The naming of each scan folder follows a certain rule. For example:

CV_P2_V_02

is the second (_02) validation scan (_V) of patient 2 (_P2) in the Clinical Varian Datasets (CV). Below is a list of the meanings of these naming characters.

Characters	Meaning
MC_/CV_/CE_	Monte Carlo/Clinical Varian/Clinical Elekta datasets
P[n]	Patient number in the particular type of datasets
_T/_V_	Training/Validation set
_Prior	Prior 4D-CT
_NS/_SC/_LD_	No scatter/With scatter/Low dose cases (see 2.6 for more details)

2.3 Prior, training and validation sets

For every patient case, the participants are provided with a XXXX_Prior folder that contains the 4D-CT images. **These are for methods that rely on using 4D-CT as a prior.**

For each type of the datasets, the participants are provided with both training (`_T_`) and validation sets (`_V_`). **The training sets are for the participants to adjust and fine-tune their algorithms, whereas algorithm performance will be assessed from the validation sets.** The validation sets contain only the down-sampled (one-minute equivalent) projection subset. The training sets contain both the down-sampled subsets and the full projection sets.

For the Monte Carlo Datasets, training sets of three patients (one scan each) are included. In contrast, for the Clinical Varian and Elekta Datasets, one training set is provided for each of the patient. This is merely due to consideration to maximize the number of patients and scans in the study. This does mean that the participants have the opportunity to fine-tune their algorithms on a patient-by-patient basis for the Clinical Varian and Elekta Datasets. **However, we encourage the participants to minimize the amount of patient-by-patient algorithm tuning, as it is generally not practically feasible.**

2.4 Files

2.4.1 Dataset information

For each type of datasets there is a `README_DataInfo.txt` file that outlines some basic imaging and geometric information and convention. This includes the dimension and spacing of projection, source-to-isocenter distance and source-to-detector distance, detector offsets, the dimension and spacing of reconstruction, and the convention of respiratory phase and bin.

2.4.2 Prior 4D-CT

Under each prior folder there are ten Metaimage files for each phase of a ten-phase 4D-CT:

`CT_01.mha, CT_02.mha, ... , CT_10.mha`

The pixel values are in Hounsfield unit.

There is also an ITV or PTV file in the folder that specifies the ITV/PTV mask.

2.4.3 Validation set

For each validation set, there are

1. Proj folder, which contains the raw projection, geometric, and respiratory information needed for 4D-CBCT reconstruction.
 - a. The Proj_XXXXX.bin files are the raw projection files.
 - b. The Geometry.xml file contains angular and geometric information for each projection.
 - c. The RespBin.csv file specifies the respiratory bin each projection belongs to for respiratory binning.
 - d. The RespPhase.csv file specifies the respiratory phase each projection corresponds to. This is for method that requires a continuous respiratory phase signal for the reconstruction.
2. FDKRecon folder, which contains example FDK reconstructions from the data in the Proj folder using the Reconstruction Toolkit developed by Dr Simon Rit¹. These reconstructions are of low image quality and are not meant for training purposes. Instead, they are provided merely as a reference to check if the participant's reconstruction is in the correct format and coordinate system. The submitted reconstruction images should be of the same format, same coordinate system, same dimension/spacing, same orientation, and similar intensity range with these FDK reconstructions. Note that the pixel values were not calibrated to Hounsfield unit, but are simply raw attenuation values as directly reconstructed from the raw projections. The participants do not need to calibrate their reconstructions to Hounsfield unit, either.
 - a. The FDK3D.mha file is the 3D-CBCT FDK reconstruction.
 - b. Each FDK4D_XX.mha file is one phase bin of the ten-phase 4D-CBCT FDK reconstruction.
 - c. The *.png and *.gif are just for quick visualization of the FDK*.mha files mentioned above.

2.4.4 Training set

For each training set, in addition to the Proj and FDKRecon folders mentioned, there are also:

¹ Reconstruction Toolkit (RTK): <https://www.openrtk.org/>

1. The Proj_Full folder that contains the original fully sampled dataset. Similarly, this folder contains the Geometry.xml, RespBin.csv, and RespPhase.csv corresponding to the each Proj_XXXXX.bin file in the folder. The participants should not use the .xml and .csv file in Proj_Full to reconstruct projections from Proj (and vice versa), as the projection numbering is different.

For the MonteCarloDatasets, projections in the Proj_Full folder are the primary signal from the simulation (excluding scatter and noise). Also, projections in the Proj_Full folder are split into ten folders and without the RespBin.csv and RespPhase.csv. The participants can simply reconstruct using all the projections in each folder separately.

2. The FDKGroundTruth folder that contains the reference reconstruction using the Proj_Full folder. Similarly, the *.mha files are the actual reconstruction files, whereas the *.png and *.gif files are for quick visualization. Note that although this folder is called the "ground truth", for MonteCarloDatasets the actual ground truths used to assess image quality are the underlying 4D-CT used in projection simulation.

2.4.5 Reading files

This section provides instructions on reading the various types of files in the datasets.

1. *.bin files

The *.bin projection files are binary files that store values of float precision (32-bit) with no header. Below is an example MATLAB script to read a 512*384 projection file:

```
>> fid = fopen('Proj_00001.bin','r');  
>> P = single(fread(fid,512*384,'single'));  
>> P = reshape(P,[512,384]);  
>> fclose(fid);
```

The participants are referred to the README_DataInfo.txt file for the projection dimension and spacing for each type of datasets.

2. *.mha files

The *.mha files can be loaded into software such as 3D Slicer². Alternatively, the MATLAB codes `mha_read_header.m` and `mha_read_volume.m` provided in MATLABCodes can be used:

```
>> header = mha_read_header('CT_01.mha');  
>> image = mha_read_volume(header);
```

PixelDimensions (i.e. pixel spacing) in header are in mm.

3. *.csv files

The *.csv files can be read as a text file, loaded in Excel, or read in MATLAB using:

```
>> data = csvread('RespBin.csv');
```

4. Geometry.xml file

This is a RTK geometry file that records the geometric information of the scan and each individual projection.

The relevant global fields (i.e. stay constant throughout the scan) are:

- `<SourceToIsocenterDistance>` (in mm)
- `<SourceToDetectorDistance>` (in mm)
- `<ProjectionOffsetX>` (not recorded if 0) (in mm)
- `<ProjectionOffsetY>` (not recorded if 0) (in mm)

kV angle for each projection is recorded under the `<GantryAngle>` tag in degrees. The MATLAB code `ReadTagFromRTKGeometry.m` can be used to extract the angle values:

```
>> angles =  
ReadTagFromRTKGeometry('Geometry.xml', 'GantryAngle');
```

For cases where `<ProjectionOffsetX>` and `<ProjectionOffsetY>` vary for each projection, they can also be extracted using `ReadTagFromRTKGeometry.m`:

² <https://www.slicer.org/>

```
>> offsetX =
ReadTagFromRTKGeometry('Geometry.xml','ProjectionOffsetX');
>> offsetY =
ReadTagFromRTKGeometry('Geometry.xml','ProjectionOffsetY');
```

To work directly with the 3D-2D transformation matrices for each projection, these are recorded under the <Matrix> tag, and can be extracted using `ReadRTKGeometryMatrices.m`:

```
>> Gs = ReadRTKGeometryMatrices('Geometry.xml');
```

The 3 by 4 geometry matrix G for each projection relates the 3D position in the patient space $(x, y, z)^T$ with the 2D position in the detector space $(u, v)^T$ through the following equation:

$$w \begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = \begin{pmatrix} G_{11} & G_{12} & G_{13} & G_{14} \\ G_{21} & G_{22} & G_{23} & G_{24} \\ G_{31} & G_{32} & G_{33} & G_{34} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix}$$

2.5 Conventions

2.5.1 Coordinate system

The 3D coordinate system for all the *.mha volumes follow the international standard IEC 61217. The x, y, and z axes correspond to left-right, superior-inferior, and anterior-posterior. The positive x direction points towards the left lung (i.e. the right side of the image). The positive y direction points towards the inferior end of the patient. The positive z direction points towards the anterior end of the patient.

The kV angles are in degrees, with 0° corresponding to when the kV source is directly above the patient and the detector is directly below the patient. Positive angular direction corresponds to clockwise rotation of the gantry when facing towards the superior end of the patient.

2.5.2 Respiratory phase and bin

Respiratory bin (1-10) for both 4D-CT and 4D-CBCT is defined such that Bin01 corresponds to end of inhale, and Bin06 corresponds to end of exhale. Similarly,

respiratory phase ($0 - 2\pi$) is defined such that 0 corresponds to end of inhale, and π corresponds to end of exhale.

2.6 Types of noise in the Monte Carlo Datasets

There are three types of noise simulation in MonteCarloDatasets.

1. For folder name with the `_NS_` tag (i.e. no scatter), the projections were a combination of **primary signals and quantum noise (tube current: 40 mA; pulse length: 20 ms)**, but **no scatter noise**.
2. For folder name with the `_SC_` tag (i.e. with scatter), the projections were a combination of **primary signals, scatter noise, and quantum noise (tube current: 40 mA; pulse length: 20 ms)**.
3. For folder name with the `_LD_` tag (i.e. low dose), the projections were a combination of **primary signals, scatter noise, and quantum noise (tube current: 20 mA; pulse length: 20 ms)**.

The different types of noise simulation allow the assessment of algorithm performance regardless of whether the participants have access to scatter correction methods. At the same time, the effect of scatter correction, if any is applied, can also be investigated. The low dose scenarios further test how far we can push with advanced algorithms.

2.7 Limitations

This section outlines the limitations with the challenge datasets.

1. For `MC_T_P2_SC`, `MC_V_P6_LD_01`, `MC_V_P6_NS_01`, `MC_V_P6_SC_01`, `MC_V_P6_SC_02`, `MC_V_P7_SC_01`, and `MC_V_P8_NS_01`, the simulated projections do not cover the entire detector in the vertical direction due to limited field-of-view of the underlying CT volumes. This will not affect the analysis, as assessment will only be based on the body region that is well within the region that can be properly reconstructed.
2. Projections in the MonteCarloDatasets may appear to have slightly lower values around the boundaries due to fluence map calibration. The impact on the reconstruction seems negligible.
3. Some of the reconstructions in `FDKGroundTruths` have slight discontinuity at the center.

4. We tried to select 4D-CT volumes with minimal artifacts and noise to simulate the projections. However, as a completely clean 4D-CT is not possible, the simulated datasets do not represent true patient anatomy perfectly.
5. Some of the prior 4D-CT volumes show noticeable motion artifacts. However, we feel that this is a realistic challenge for prior based methods, as 4D-CT artifacts are quite common in clinical practice.
6. The respiratory phase and bin information for the `MonteCarloDatasets` are discrete, because the projections were simulated using ten 4D-CT volumes instead of continuously changing anatomy. This means that there is zero motion within each respiratory bin, which is unrealistic. However, this also allows the assessment to purely focus on the effect of reconstruction algorithms and not affected by binning related issues.
7. The respiratory phases and bins for the `ClinicalVarianDatasets` and `ClinicalElektaDatasets`, which were acquired from the RPM system and Amsterdam Shroud method, respectively, may not be the perfect binning signal for some cases. This is due to issues related with external-internal correlation (RPM) and image quality (Amsterdam Shroud). However, we ask the participants to perform their 4D reconstructions using the exact phase and binning information provided. This way we ensure differences between reconstruction results are solely due to the differences in algorithms.
8. The geometry information for some of the `ClinicalElektaDatasets` may not be 100% accurate. This is because some of the original geometry files were lost on the clinical systems. In those cases, the provided `Geometry.xml` files were estimated by assuming even angular increment, which seems to be reasonable. We ask the participants to use the exact geometry information provided to ensure differences in reconstruction results are solely due to differences in algorithms.

3 Reconstruction and naming

This section outlines how the participants are expected to perform the reconstructions and name the files in a way convenient for evaluation.

3.1 Folder structure and reconstruction files

The reconstruction results should be saved under a standalone folder (e.g. "ReconResults") with the exact same folder structure and folder naming as provided in the participant datasets. Under each scan folder, there should only be ten files that represent the ten-phase reconstruction following the numbering convention mentioned in 2.5.2:

Recon_01.mha, Recon_02.mha,, Recon_10.mha

3.2 File format and coordinate system

The reconstructed volumes should be saved as uncompressed 32-bit (float) Metaimage (.mha) format.

The coordinate system should follow the international standard IEC 61217. The x, y, and z axes correspond to left-right, superior-inferior, and anterior-posterior. The positive x direction points towards the left lung (i.e. the right side of the image). The positive y direction points towards the inferior end of the patient. The positive z direction points towards the anterior end of the patient.

Participants do not need to calibrate the pixel values to Hounsfield unit. The reconstructed volumes should be obtained by directly applying the reconstruction algorithms to the raw projections without scaling. The pixel values should range roughly around 0-0.008 for lungs, 0.01-0.015 for soft tissues, and 0.016-0.03 for bony anatomy.

3.3 Reconstruction dimension and voxel size

The reconstruction voxel size should be 1 mm in all directions. While this is smaller than what is clinically used, this allows the assessment to be performed at a finer level.

Reconstruction dimension should be [450,220,450] (LR,SI,AP) for MonteCarloDatasets and ClinicalVarianDatasets, and [270,256,270] for

ClinicalElektaDatasets. This matches with the reconstruction field-of-view. This information is also recorded in the README_DataInfo.txt file.

3.4 Projection dimension and spacing

The dimension and pixel spacing for the projections are recorded in the README_DataInfo.txt file for each type of datasets.

3.5 Using prior 4D-CT, respiratory and geometry information

The participants are allowed to use the prior 4D-CT in any way to assist the reconstructions.

Please use the respiratory phase, bin, and scan geometry exactly as provided to ensure all reconstructions are computed using the same information.

3.6 Scatter and projection correction

The participants can apply any form of scatter correction, but are not required to.

We are aware that many of the participants may not have access to scatter correction. Thus, in the MonteCarloDataets, projections with or without scatter noise were both included. This way, algorithm performance regardless of scatter correction can be assessed based on the datasets without scatter, while the impact of scatter correction can be investigated by comparing results on datasets with and without scatter.

Please do not apply additional corrections regarding scan geometry (e.g. additional detector shift not recorded in Geometry.xml) so that differences in the reconstruction results are solely due to the ability of the algorithms to deal with data sparsity.

3.7 Compliance check

The participants should use the reference reconstructions in the FDKRecon folder to check compliance in terms of file format, coordinate system, and intensity range.

4 Evaluation

Once the reconstruction results are generated and saved according to the previous section, quantitative analysis can be automatically calculated using the provided MATLAB scripts. Briefly, the quantitative analysis investigates the similarity of the reconstruction with the ground truth in terms of RMSE and SSIM, as well as its geometric accuracy in terms of the position of the PTV. For more details, please refer to the SPARE Challenge paper. Note also that the quantitative analysis is only available with the Monte Carlo Datasets due to the lack of ground truth in the Clinical Varian and Clinical Elekta datasets.

To compute the quantitative analysis:

1. All files in SPARE_EvaluationCodes.7z should be uncompressed and saved in one single folder.
2. All files in SPARE_GroundTruth.7z should be uncompressed and saved under the same folder in step 1. The final folder structure should be something like this:

Evaluation

```
MonteCarloDatasets
ClinicalVarianDatasets
ClinicalElektaDatasets
spare_BatchRunImageAnalysis.m
spare_BatchRunGeometricAnalysis.m
.....
```

3. In MATLAB, navigate to the folder.
4. Run the following commands to perform image analysis (similarity with ground truth) and geometric analysis (PTV accuracy):

```
>> spare_BatchRunImageAnalysis(YourFolderPath);
>> spare_BatchRunGeometricAnalysis(YourFolderPath);
```

YourFolderPath is a string variable storing the path to the top folder where you saved your reconstructions.

The calculation will likely take a day or even longer.

5. To display the summary of the analysis, run:

```
>> summary = spare_GetResultSummary;
```

Please refer to the README.txt file included in SPARE_EvaluationCodes.7z for more information.

Users of the datasets are welcome to compare their results directly with the top 5 methods as presented in the SPARE Challenge paper.